

Start coding or [generate](#) with AI.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
import pandas as pd

df = pd.read_excel("/Excel.xlsx")

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Product ID	Category	C
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	FUR-BO-10001798	Furniture	Bc
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	FUR-CH-10000454	Furniture	
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	OFF-LA-10000240	Office Supplies	
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	FUR-TA-10000577	Furniture	
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	OFF-ST-10000760	Office Supplies	

5 rows × 23 columns

```
# Convert dates
df['Order Date'] = pd.to_datetime(df['Order Date'])

# KPIs
print("Total Sales:", df['Sales'].sum())
print("Total Profit:", df['Profit'].sum())
print("Total Orders:", df['Order ID'].nunique())

# Region-wise
region = df.groupby('Region').agg({
    'Sales': 'sum',
    'Profit': 'sum'
})

print(region)
```

```
Total Sales: 2297200.8603000003
Total Profit: 286397.0217
Total Orders: 5009
      Sales      Profit
Region
Central 501239.8908  39706.3625
East    678781.2400  91522.7800
South   391721.9050  46749.4303
West    725457.8245  108418.4489
```

```
df.to_csv("cleaned_data.csv", index=False)
```

```
# Create Year column
df['Year'] = df['Order Date'].dt.year
```

```
# Create Month column
df['Month'] = df['Order Date'].dt.month_name()

# Create Profit Margin
df['Profit Margin'] = df['Profit'] / df['Sales']

df.head()
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...	Category	Sub-Category	Product
0	1	CA-2016-152156	2016-11-08	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	Furniture	Bookcases	Son Coll. Boo
1	2	CA-2016-152156	2016-11-08	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...	Furniture	Chairs	Hon D I Upholste Sta Ch
2	3	CA-2016-138688	2016-06-12	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...	Office Supplies	Labels	Adf Ad Lab Typev
3	4	US-2015-108966	2015-10-11	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	Furniture	Tables	Br CF Series Rectan
4	5	US-2015-108966	2015-10-11	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...	Office Supplies	Storage	Eldor 'N Ro S

5 rows × 24 columns

```
region_perf = df.groupby('Region').agg({
    'Sales': 'sum',
    'Profit': 'sum',
    'Order ID': 'nunique'
}).reset_index()

region_perf.columns = ['Region', 'Total Sales', 'Total Profit', 'Total Orders']

region_perf
```

	Region	Total Sales	Total Profit	Total Orders
0	Central	501239.8908	39706.3625	1175
1	East	678781.2400	91522.7800	1401
2	South	391721.9050	46749.4303	822
3	West	725457.8245	108418.4489	1611

```
category_perf = df.groupby('Category').agg({
    'Sales': 'sum',
    'Profit': 'sum'
}).reset_index()

category_perf
```

	Category	Sales	Profit
0	Furniture	741999.7953	18451.2728
1	Office Supplies	719047.0320	122490.8008
2	Technology	836154.0330	145454.9481

```
monthly_sales = df.groupby('Month')['Sales'].sum().reset_index()

monthly_sales
```

	Month	Sales
0	April	137762.1286
1	August	159044.0630
2	December	325293.5035
3	February	59751.2514
4	January	94924.8356
5	July	147238.0970
6	June	152718.6793
7	March	205005.4888
8	May	155028.8117
9	November	352461.0710
10	October	200322.9847
11	September	307649.9457

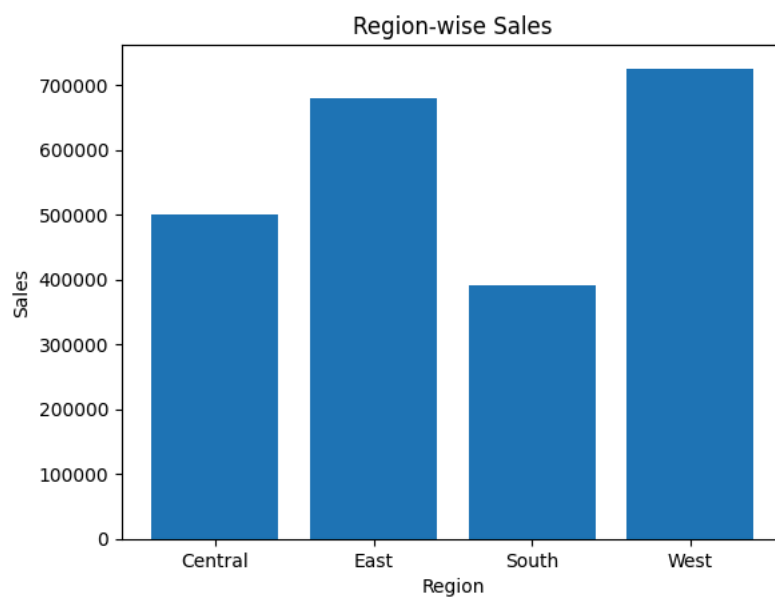
```
segment_perf = df.groupby('Segment').agg({  
    'Sales': 'sum',  
    'Profit': 'sum'  
}).reset_index()
```

segment_perf

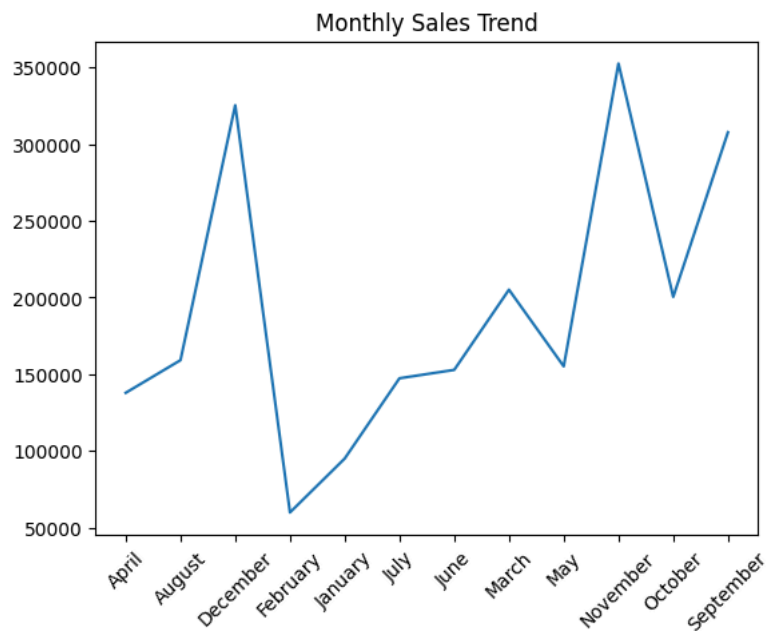
	Segment	Sales	Profit
0	Consumer	1.161401e+06	134119.2092
1	Corporate	7.061464e+05	91979.1340
2	Home Office	4.296531e+05	60298.6785

```
import matplotlib.pyplot as plt
```

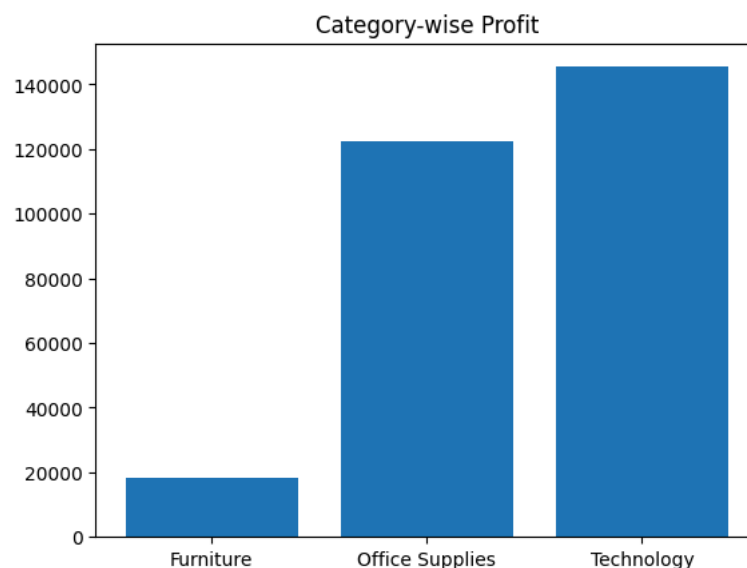
```
plt.figure()  
plt.bar(region_perf['Region'], region_perf['Total Sales'])  
plt.title("Region-wise Sales")  
plt.xlabel("Region")  
plt.ylabel("Sales")  
plt.show()
```



```
plt.figure()  
plt.plot(monthly_sales['Month'], monthly_sales['Sales'])  
plt.title("Monthly Sales Trend")  
plt.xticks(rotation=45)  
plt.show()
```



```
plt.figure()
plt.bar(category_perf['Category'], category_perf['Profit'])
plt.title("Category-wise Profit")
plt.show()
```



```
def generate_report(df):
    print("Total Sales:", df['Sales'].sum())
    print("Total Profit:", df['Profit'].sum())
    print("Total Orders:", df['Order ID'].nunique())

    print("\nRegion Performance:\n", region_perf)
    print("\nCategory Performance:\n", category_perf)

generate_report(df)
```

```
Total Sales: 2297200.8603000003
Total Profit: 286397.0217
Total Orders: 5009
```

Region Performance:

	Region	Total Sales	Total Profit	Total Orders
0	Central	501239.8908	39706.3625	1175
1	East	678781.2400	91522.7800	1401
2	South	391721.9050	46749.4303	822
3	West	725457.8245	108418.4489	1611

Category Performance:

	Category	Sales	Profit
0	Furniture	741999.7953	18451.2728
1	Office Supplies	719047.0320	122490.8008
2	Technology	836154.0330	145454.9481

```
df.to_csv("cleaned_data.csv", index=False)
region_perf.to_csv("region_summary.csv", index=False)
category_perf.to_csv("category_summary.csv", index=False)
```

Start coding or [generate](#) with AI.